

WINDOWS SUBSYSTEM FOR LINUX (WSL2)
& VISUAL STUDIO CODE
INTEGRATION SETUP GUIDE

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1. INTRODUCTION

1.1 Purpose

This document provides comprehensive instructions for installing and configuring Windows Subsystem for Linux 2 (WSL2) with Visual Studio Code integration, enabling native Linux development on Windows platforms.

1.2 Scope

- WSL2 installation and configuration
- Ubuntu Linux distribution setup
- Visual Studio Code installation
- WSL-VS Code integration
- Development environment verification

1.3 Target Audience

- Software developers
- DevOps engineers
- IT administrators
- Technical staff requiring Linux environment on Windows.

2. PREREQUISITES

Before beginning installation, ensure the following requirements are met:

- ☐ Windows 10 (Version 2004 / Build 19041 or higher)
- ☐ Windows 11 (any version)
- ☐ Administrator privileges on the machine
- ☐ Hardware virtualization enabled in BIOS/UEFI
- ☐ Minimum 4GB RAM (8GB recommended)
- ☐ At least 10GB free disk space
- ☐ Active internet connection
- ☐ PowerShell 5.1 or higher.

3. INSTALLING WSL2

3.1 Enable WSL Feature

Step 1: Open PowerShell as Administrator

- Right-click Start menu
- Select "Windows PowerShell (Admin)" or "Terminal (Admin)"

Step 2: Execute installation command:

```
wsl --install
```

Step 3: Restart your computer when prompted

3.2 Initial Ubuntu Setup

After restart, a terminal window will automatically open:

- Enter UNIX username (lowercase letters, no spaces)
- Create password (input will be hidden - type blindly)
- Confirm password

3.3 Verify Installation

Open PowerShell and run:

```
wsl -l -v
```

Expected output:

NAME	STATE	VERSION
Ubuntu	Running	2

✓ Confirm VERSION shows "2"

3.4 Upgrade to WSL2 (if needed)

If version shows "1", upgrade with:

```
wsl --set-version Ubuntu 2
```

Update WSL kernel:

```
wsl --update
```

4. CONFIGURING LINUX ENVIRONMENT

4.1 Launch Ubuntu

- Click Start menu
- Search for "Ubuntu"
- Click to launch

4.2 Update System Packages

Execute the following commands:

```
sudo apt update  
sudo apt upgrade -y
```

4.3 Install Development Tools (Optional)

```
sudo apt install -y build-essential git curl wget unzip
```

This installs:

- build-essential: C/C++ compiler and build tools
- git: Version control system
- curl: Data transfer utility
- wget: File download utility
- unzip: Archive extraction tool

4.4 Verify WSL2 Kernel

```
uname -r
```

Expected output contains: microsoft-standard-WSL2

4.5 Configure Default User (Optional)

To set default user in wsl.conf:

```
sudo nano /etc/wsl.conf
```

Add:

```
[user]  
default=<your-username>
```

Save (Ctrl+O, Enter) and exit (Ctrl+X)

5. INSTALLING VISUAL STUDIO CODE

5.1 Download VS Code

Visit official website:

<https://code.visualstudio.com/>

Click "Download for Windows"

5.2 Installation Steps

Step 1: Run VSCodeSetup-x.x.x.exe

Step 2: Accept license agreement

Step 3: Select installation options:

- ☒ Add to PATH (required)
- ☒ Register Code as editor for supported file types
- ☒ Add "Open with Code" to context menu
- ☒ Create desktop icon (optional)

Step 4: Click "Install"

Step 5: Click "Finish" when complete

5.3 Initial Launch

- Launch Visual Studio Code
- Accept license terms
- Choose color theme (optional)
- Close welcome screen

6. CONNECTING VS CODE TO WSL

6.1 Method A: From Ubuntu Terminal (Recommended)

Step 1: Open Ubuntu terminal

Step 2: Navigate to project directory:

```
cd ~/projects/my-app
```

Step 3: Launch VS Code:

```
code .
```

First-time setup:

- VS Code will install server components in WSL
- Accept installation prompts
- Wait for "WSL: Server ready" notification

6.2 Method B: From VS Code Interface

Step 1: Open VS Code

Step 2: Install WSL Extension:

- Press Ctrl+Shift+X
- Search "WSL"
- Install "WSL" by Microsoft

Step 3: Connect to WSL:

- Click green remote indicator (bottom-left)
- Select "Connect to WSL"
- Choose Ubuntu distribution

6.3 Verify Connection

- Status bar shows: "WSL: Ubuntu"
- File explorer shows Linux filesystem
- Terminal opens in Linux environment

7. VERIFICATION & BEST PRACTICES

7.1 Verification Checklist

Test	Command/Action	Expected Result
Terminal shell	Ctrl+`	Linux prompt (\$)
Current directory	pwd	/home/username/...
Linux → Windows access	explorer.exe .	Windows Explorer opens
Windows → Linux access	\\wsl.localhost\ in path	Linux files accessible
VS Code remote	Bottom-left indicator	Shows "WSL: Ubuntu"

7.2 Best Practices

DO:

- ✓ Store projects in Linux filesystem (/home/username/)
- ✓ Use VS Code integrated terminal for commands
- ✓ Keep WSL and extensions updated
- ✓ Use Linux-native tools and compilers
- ✓ Regular system updates: `sudo apt update && sudo apt upgrade`

DON'T:

- X Edit Linux files with Windows applications

- X Store projects on Windows drives (/mnt/c/)
- X Run VS Code as Administrator
- X Mix Windows and Linux line endings carelessly
- X Ignore permission errors

7.3 File System Access

Linux to Windows:

- Windows drives mounted at: /mnt/c/, /mnt/d/, etc.
- Access: `cd /mnt/c/Users/YourName`

Windows to Linux:

- Network path: `\\wsl.localhost\Ubuntu\home\username`
- Or: `\\wsl$\Ubuntu\home\username`

8. TROUBLESHOOTING

Problem: `wsl --install` fails or errors

Solution:

1. Enable virtualization in BIOS/UEFI
2. Update Windows to latest version
3. Manually enable features:

```
dism.exe /online /enable-feature /featurename:Microsoft-Windows-Subsystem-Linux /all /norestart
```

```
dism.exe /online /enable-feature /featurename:VirtualMachinePlatform /all /norestart
```

4. Restart computer

5. Download WSL2 kernel update:

<https://aka.ms/wsl2kernel>

Problem: WSL stays on version 1

Solution:

1. Ensure Virtual Machine Platform is enabled

2. Set WSL2 as default:

```
wsl --set-default-version 2
```

3. Convert existing distro:

```
wsl --set-version Ubuntu 2
```

Problem: "code: command not found" in WSL

Solution:

1. Reinstall VS Code with "Add to PATH" option

2. Run VS Code once from Windows

3. In WSL, add to .bashrc:

```
echo 'export PATH="$PATH:/mnt/c/Users/YourName/AppData/Local/Programs/Microsoft VS Code/bin"' >> ~/.bashrc
```

```
source ~/.bashrc
```

Problem: VS Code won't connect to WSL

Solution:

1. Shutdown WSL:

```
wsl --shutdown
```

2. Restart VS Code
3. Reinstall WSL extension
4. Check WSL service is running:

```
Get-Service -Name LxssManager
```

Problem: Permission denied errors

Solution:

1. Fix ownership:

```
sudo chown -R $USER:$USER ~/your-project
```

2. Avoid running VS Code as Administrator
3. Check file permissions:

```
ls -la ~/your-project
```

9. APPENDIX: USEFUL COMMANDS

WSL Management Commands:

List installed distributions:

```
wsl -l -v
```

Start specific distribution:

```
wsl -d Ubuntu
```

Stop all WSL instances:

```
wsl --shutdown
```

Set default distribution:

```
wsl --set-default Ubuntu
```

Export distribution:

```
wsl --export Ubuntu ubuntu-backup.tar
```

Import distribution:

```
wsl --import Ubuntu C:\WSL\Ubuntu ubuntu-backup.tar
```

Uninstall distribution:

```
wsl --unregister Ubuntu
```

Linux Commands:

Update system:

```
sudo apt update && sudo apt upgrade -y
```

Install package:

```
sudo apt install <package-name>
```

Remove package:

```
sudo apt remove <package-name>
```

Check disk usage:

```
df -h
```

Check memory usage:

`free -h`

View processes:

`top`

File Navigation:

Change directory:

`cd /path/to/directory`

List files:

`ls -la`

Create directory:

`mkdir dirname`

Copy files:

`cp source destination`

Move/rename files:

`mv source destination`

Delete files:

`rm filename`

VS Code Shortcuts:

Open terminal: `Ctrl+``

Command palette: `Ctrl+Shift+P`

Quick open:	Ctrl+P
Extensions:	Ctrl+Shift+X
Settings:	Ctrl+,
Toggle sidebar:	Ctrl+B
Split editor:	Ctrl+\